Contents

HerHealthEval — Baseline & Full Results Report (in depth)

Study: *Do language models understand women's-health communication across languages and cultures?* **Object of study:** language *understanding* (interpretation, triage calibration, clarification, cross-language/register consistency) — **not** answer generation. **Prepared for supervisor review, to serve as the empirical basis for the paper.**

All numbers are computed deterministically by the committed evaluators (`scripts/evaluate.py`, `scripts/safety_metrics.py`, `scripts/multilingual_report.py`) on the aligned benchmark `benchmark_multilingual_v1`, N = 540 items per language. Every figure in this report is reproducible with one command (see §9).

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1. Executive summary

- The **baseline** is **M2 = base Qwen3.5-9B, zero-shot** (no fine-tuning), prompted with our structured triage schema. It is *also* the multilingual reference, because Qwen3.5-9B is natively multilingual.
- The baseline **interprets** the concern (category accuracy) at **0.617 / 0.614 / 0.617** in EN / FR / AR — remarkably **stable across languages** (well above the 0.333 majority baseline).
- Its safety-critical weakness is **under-triage** — routing a *see-doctor* case to *routine* — at **0.463 / 0.440 / 0.420** (EN/FR/AR): the base model misses ~42–46% of cases that should see a clinician, but does so *consistently* across languages.
- Two strong structure effects the paper is built on:
  1. **Register (surface form) drives interpretation.** Interpretation falls from ~0.66–0.68 on *clinical* phrasing to ~0.46–0.51 on *ambiguous* phrasing — a ~20-point swing on the **same underlying case**. Cross-style consistency is only **0.433** for risk (the model changes its risk verdict across the six phrasings of a case more than half the time).
  2. **Condition category matters enormously.** Fertility items are interpreted at ~0.89–0.92 but menstrual items at only ~0.39–0.40 — a ~50-point gap that is stable across all three languages.
- **Cross-language consistency** is genuine: for the same seed, the base model gives the same category verdict across EN/FR/AR **82.4%** of the time and the same risk verdict **71.1%** of the time — interpretation is largely language-invariant, and the residual disagreement is the honest cross-lingual instability the benchmark is designed to expose.
- The baseline retains **non-trivial clarification behaviour** (recall 0.21–0.38 on genuinely ambiguous items) — a capability both fine-tunes lose entirely.

Why this matters for the paper. The baseline establishes that a strong, natively-multilingual model (i) understands the concern moderately well and language-invariantly, yet (ii) is surface-form-sensitive and (iii) unsafe on ~44% of high-risk cases. Everything else in the study — how English and multilingual fine-tuning move these numbers — is measured *against this reference.*

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2. The benchmark (`benchmark_multilingual_v1`)

Property	Value
Seed clinical conditions	90
Categories (balanced)	menstrual, PCOS, fertility → 180 items each per language
Communication registers	canonical, clinical, layperson, indirect_cultural, ambiguous, emotionally_concerned (6)
Items per language	90 seeds × 6 registers = **540**
Languages	English, French, Arabic → **1,620 parallel items**
Gold labels	concern **category**, **risk level**, recommended **action**, **requires-clarification**

Parallel design. Each `menst-XXX` (etc.) seed appears in all 3 languages × 6 registers with the *same* gold labels — labels are a property of the underlying case, not the surface form. This is what makes cross-language consistency and cross-register consistency directly measurable.

Gold-label distribution (identical across languages by design): - **risk_level: 100% see-doctor** → risk majority baseline = 1.0. The honest headline is therefore **under-triage rate**, not risk accuracy. - **requires_clarification: 24 yes / 516 no** → clarification recall is measured on the 24 genuinely-underspecified (ambiguous) items; specificity on the other 516. - **category: balanced 180/180/180** → category (interpretation) accuracy is informative, majority baseline 0.333.

Translation-validation protocol. GPT-assisted initial translation → (1) native-speaker review for fluency/naturalness → (2) linguistic-professional review for meaning preservation, ambiguity, and cultural appropriateness. Per-item validation status is carried in the benchmark.

Benchmark hygiene (important). An earlier English-only set (`gss-*` seeds) is a *different* question set (2/540 text overlap) with different clarification labels. It is **excluded** from all multilingual claims; the report's seed-namespace guard drops it automatically. (This is why M3ml-v1 has no aligned EN row.)

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3. Metric definitions

All metrics are computed from the model's structured JSON prediction against gold.

Metric	Definition	Notes
parse_ok	fraction of responses that parse into the required schema	format compliance
Interpretation accuracy	predicted concern category == gold category	majority baseline 0.333
Under-triage rate ↓	P(predict `routine` | gold `see-doctor`)	the safety-critical error; lower is better
Clarification recall	of the 24 gold "needs clarification" items, fraction where the model asks	measured only on genuine ambiguity

Metric	Definition	Notes
Clarification specificity	of the 516 "no clarification" items, fraction where the model does *not* ask	guards against always-asking
Indirect-cultural interpretation	interpretation accuracy restricted to the `indirect_cultural` register	cultural-sensitivity proxy
Cross-language consistency	for a fixed seed, fraction of seeds where the verdict agrees across EN/FR/AR	meaning preservation
Cross-style consistency	for a fixed seed, fraction of seeds where the verdict is identical across all 6 registers	surface-form robustness

Statistics. Bootstrap 95% CIs on rates (2,000 resamples); **McNemar's paired test** on shared items for model-vs-model comparisons (reported later). Because gold risk is uniformly `see-doctor`, we always report under-triage and per-class behaviour rather than a single risk-accuracy number.

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4. Baseline (M2) results — in depth

4.1 Overall, per language (with 95% CIs)

Lang	parse_ok	interpretation [95% CI]	under-triage ↓ [95% CI]	clar. recall	clar. spec
EN	1.000	0.617 [0.576, 0.657]	0.463 [0.422, 0.506]	0.208	0.878
FR	0.998	0.614 [0.573, 0.655]	0.440 [0.399, 0.482]	0.333	0.862
AR	1.000	0.617 [0.574, 0.656]	0.420 [0.380, 0.463]	0.375	0.849

Reading: interpretation is statistically identical across the three languages (overlapping CIs); the base model understands the concern equally well regardless of language. Under-triage is high everywhere (~0.42–0.46) but, again, consistent.

4.2 By communication register (the surface-form effect)

Interpretation accuracy / under-triage per register (averaged behaviour; each cell n≈90):

Register	EN interp	FR interp	AR interp	EN u-triage	FR u-triage	AR u-triage
canonical	0.667	0.652	0.644	0.444	0.416	0.444
clinical	0.678	0.667	0.667	0.478	0.522	0.467
layperson	0.644	0.611	0.611	0.489	0.433	0.433
indirect_cultural	0.611	0.600	0.611	0.478	0.389	0.367
ambiguous	**0.456**	**0.511**	**0.500**	0.533	0.500	0.478
emotionally_concerned	0.644	0.644	0.667	0.356	0.378	0.333

Key finding (RQ1). Holding the clinical case fixed, moving from *clinical* to *ambiguous* phrasing drops interpretation by ~20 points in every language, and *ambiguous* also carries the highest under-triage. Indirect/cultural phrasing costs a smaller but consistent ~5–7 points. The model's understanding is **driven by surface form, not just clinical content**.

4.3 By condition category (a large, stable disparity)

Interpretation accuracy per category (n≈180 each):

Category	EN	FR	AR
fertility	0.917	0.894	0.889
PCOS	0.539	0.553	0.561
menstrual	**0.394**	**0.394**	**0.400**

Key finding. The base model recognises fertility concerns almost perfectly but menstrual concerns poorly (~0.39), with PCOS in between — a ~50-point gap that holds across all three languages.

Error analysis (see `error_analysis_menstrual.md`) shows this is largely a category-boundary artifact, not a blind spot. Of the wrong menstrual items, **87% are mapped to an adjacent category** (fertility or PCOS) and only 13% to "other". The reason: **14 of the 30 menstrual seeds (47%) explicitly concern conception/pregnancy** (e.g. *"Do irregular periods influence the ability to get pregnant?"*, *"trying for child since 1 yr"*) — labeled `menstrual` by presenting symptom but *fertility* by patient intent, which is what the model surfaces. One seed (irregular periods + polycystic ovaries on ultrasound) is a textbook PCOS presentation the model arguably reads *more* correctly than the gold. The confusion is stable across all six registers and all three languages, and fine-tuning merely shifts it (base → fertility; fine-tunes → PCOS) without resolving it. **Recommendation:** report a relaxed clinically-acceptable interpretation metric and/or refine the conception-motivated menstrual labels, and frame the gap as *category-boundary ambiguity in real patient language* — a substantive finding rather than a weakness.

4.4 Cross-style consistency (same case, six phrasings)

Lang	risk consistency	category consistency
EN	0.433	0.489
FR	0.433	0.478
AR	0.433	0.522

Reading. For only ~43% of seeds does the base model give the *same* risk verdict across all six phrasings, and ~half for category. This is direct evidence that interpretation is unstable under paraphrase — the central motivation for a register-controlled benchmark.

4.5 Cross-language consistency (same seed, three languages)

Risk **0.711**, category **0.824** (n = 540 aligned triples). Interpretation is largely language-invariant (82% agreement); the residual ~18% category and ~29% risk disagreement is genuine cross-lingual instability over a *live* risk distribution — not an artifact.

4.6 Clarification behaviour

The base model **does** ask for clarification on genuinely ambiguous input — recall 0.21 (EN) rising to 0.33–0.38 (FR/AR) — at high specificity (0.85–0.88, i.e. it rarely over-asks). This calibrated "ask when unsure" capability is a baseline strength; §6 shows both fine-tunes destroy it.

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5. Full study context — how the fine-tunes move the baseline

The baseline is the reference for a two-step adaptation study.

Model	EN interp / u-triage	FR interp / u-triage	AR interp / u-triage
M2 (base)	0.617 / 0.463	0.614 / 0.440	0.617 / 0.420
M3ml-v1 (buggy labels)	— (legacy set)	0.605 / **1.000**	0.618 / **0.998**
M3ml-v2 (corrected)	**0.644** / 0.439	0.617 / 0.450	**0.646** / 0.443

- **M3ml-v1** = QLoRA multilingual fine-tune whose risk labels were derived by an English consult-word heuristic run on *translated* answers, mislabeling 99.8% of FR/AR training rows `routine`. Result: it understands FR/AR as well as the base model but **under-triages ~100%** of high-risk cases — i.e. naive multilingual fine-tuning made the model *less safe than the baseline*.
- **M3ml-v2** = identical recipe with risk labels recovered language-independently from the English source by `row_id`. Result: under-triage returns to **0.45/0.44 ≈ baseline**, and interpretation becomes the **best of any configuration** in every language.

Statistical confirmation (McNemar, paired, shared 540 items; b = first-model right & second wrong, c = reverse)

Comparison	Lang	Measure	b	c	p
M2 vs M3ml-v1	FR	risk	289	0	2.2e-64
M2 vs M3ml-v1	AR	risk	298	0	2.5e-66
M3ml-v1 vs M3ml-v2	FR	risk	0	292	5.0e-65
M3ml-v1 vs M3ml-v2	AR	risk	0	296	6.7e-66
M2 vs M3ml-v2	FR	risk	84	87	0.88
M2 vs M3ml-v2	AR	risk	83	81	0.94
M2 vs M3ml-v2	EN	risk	63	96	0.011
(all)	all	category	—	—	n.s. (0.06–1.0)

Interpretation: the base-vs-v1 risk discordance is total and one-directional (c = 0): the baseline is safer wherever they disagree. The correction (v1→v2) flips it completely (b = 0). After the fix, v2 vs the baseline is statistically **indistinguishable** on risk in FR/AR (p = 0.88/0.94) and slightly safer in EN — i.e. corrected supervision restored baseline safety — while category interpretation is unchanged throughout. **Clarification** is the exception: both fine-tunes drop to 0.000 recall, so the baseline retains a real advantage there (p ~ 1e-11).

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6. What the baseline tells us (for writing the paper)

1. **A strong multilingual base model is only moderately safe out of the box** (~44% under-triage on curated high-risk items) — motivating the whole "understanding ≠ safety" framing.
2. **Interpretation is language-invariant but form-sensitive.** Cross-language consistency is high (0.82 category) while cross-style consistency is low (0.43 risk). This dissociation is the paper's cleanest empirical hook for RQ1/RQ2.
3. **Category and register are the two axes of failure**: menstrual concerns and ambiguous phrasing are where interpretation collapses. Both are stable across languages, so they generalise.
4. **The baseline sets the safety bar that fine-tuning must not fall below** — and the study's headline is precisely that naive fine-tuning *did* fall below it, and corrected supervision climbed back to it.
5. **Clarification is the open problem** at every stage (baseline modest, fine-tunes zero) — the clearest direction for future work.

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7. Threats to validity / honest limitations

- **Gold risk is uniformly `see-doctor`** (curated high-stakes set). This supports under-triage analysis but not full triage calibration (no `routine`/`urgent` ground truth to score specificity of the risk head).

- **Cultural sensitivity is proxied** by `indirect_cultural`-register performance, not by human cultural-appropriateness ratings.
- **Response helpfulness/clarity** is not yet automatically scored (LLM-judge candidate).
- The **menstrual interpretation gap** needs a qualitative error analysis before strong claims (is it a labeling artifact or a genuine model blind spot?).
- Single fine-tuning seed (3407); no multi-seed variance yet.

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8. Suggested paper structure grounded in these results

1. **Framing:** understanding $\neq$ safe answering; register + language as controlled axes.
2. **Benchmark contribution:** the parallel EN/FR/AR $\times$ 6-register design (§2).
3. **Baseline analysis (this report):** §4 — language-invariance, register sensitivity, category disparity, cross-style instability.
4. **Adaptation study:** §5 — the v1 safety regression and its v2 correction, with the McNemar ablation isolating *label quality* as the mechanism.
5. **Error analysis:** menstrual gap + ambiguous-register collapse + clarification loss.
6. **Limitations & ethics:** §7.

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9. Reproducibility

Every number here regenerates from committed code and data:

```
# headline table + cross-language consistency + all McNemar p-values
python scripts/multilingual_report.py \
  --dir Used_Datasets/Consolidated_Datasets/200_Seed_Dataset \
  --model M2ml --model M3ml --model M3ml_v2 --langs en,fr,ar --latex
```

Raw predictions: `result/predictions/*.jsonl` (also under `Used_Datasets/Consolidated_Datasets/200_Seed_Dataset/`). Per-register / per-category / cross-style / CI breakdowns in §4 were computed with `scripts/evaluate.py` + `scripts/safety_metrics.py` scorers over those same files.

All artifacts are on branch `hassan-pc`. The compiled paper is `result/paper/HerHealthEval.pdf`; the living results doc is `result/HerHealthEval_methodology_results.md`.